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(1) Research Program/Project/Study Title: Performance of Purple Pechay (<i>Brassica rapa</i> var. <i>chinensis</i>) under Hydroponic Culture using Organic Substrate	
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Program or Project has component (if no proceed to 2)? ☐ Yes ☐ No

Component Title:	
Component Leader:	
Academic Rank:	

(2) Category (please check) ☐ Social Science/Education /Humanities & other related fields ☒ Natural/Biological Sciences ☐ Food Research/ ☐ Developmental/ Computer/Engineering/ Architecture ☐ Extension ☐ Poster	(3) Research Center [Check the concerned center] ☐ SSITRDC Social Science and Industrial Technology Research Development Center ☐ LRDC Livestock Research Development Center ☐ FFRDC Fisheries and Food Research Development Center ☒ CSRDC Crop Science Research Development Center
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(4) Site of implementation			
Municipality/City:		Mambusao	
Province:		Capiz	
Region:		VI	
(5) Classification Research: Development: ☐ Basic ☐ Pilot Testing ☒ Applied ☐ Tech. Promotion/Commercialization		(6) Mode of Implementation ☒ Single Authorship/Proponent ☐ Collaborative Research (within campus) ☐ Collaborative Research (intercampus) ☐ Collaborative Research (external agency)	
(7) Sector/Commodity Purple Pechay		(8) Discipline/Field of Specialization Agriculture	

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(9A) Abstract (This has a maximum of 250 words. It contains 1-2 sentences of rationale, followed by objectives, brief methodology, major findings, and implications of impact of the study.)

The study was conducted at the Organic Agriculture Project of CAPSU Burias, Mambusao, Capiz, Philippines. The experiment was laid out using the Completely Randomized Design, replicated three times. The treatments consisted of different substrates: A – SNAP, B – CAFSA with chicken dung and molasses, C – CAFSA with cow manure without molasses, D – CAFSA with chicken dung without molasses, and E – CAFSA with hog manure without molasses. Growth and yield parameters were gathered and analyzed using the Statistical Tool for Agricultural Research (STAR) software, except for the descriptive data. Differences among treatment means were compared using the Least Significant Difference test (LSD).

Results showed that the growth of purple pechay, as indicated by plant height, stem girth, number of leaves, width and length of leaves, length of roots, and root weight, was influenced variably by the hydroponic substrates. The yield of purple pechay, as indicated by biomass and weight of marketable parts, was significantly influenced by the hydroponic substrates. Among the organic substrates, CAFSA with cow manure without molasses performed better with higher plant height wider leaves, longer leaves, and longer roots, heavier biomass and weight of marketable part.

(9B) Keywords (Write 3-6 words you used in the study, preferably not found in the title because words in the title are already given keywords. This would also include process, or methodology employed.)

Lettuce, Hydroponic Culture, Organic, CAFSA

(10) Introduction (This includes the rationale, background of the study, and objectives which are not divided by subheadings. Incorporate at least three (3) reviews)

Purple Pechay (*Brassica rapa var. chinensis*), also known as Chinese cabbage, is a leafy vegetable gaining popularity due to its nutritional benefits and vibrant coloration, which is rich in anthocyanins. Hydroponic systems, which allow plants to grow in nutrient solutions without soil, are becoming increasingly popular for their ability to enhance growth rates, yield, and nutrient content of various crops (Jones, 2005).

Hydroponic cultivation has been demonstrated to provide several advantages over traditional soil-based agriculture. These advantages include increased control over nutrient supply, reduced incidence of soil-borne diseases, and the potential for higher yield and quality of produce (Resh, 2012). Organic substrates, such as coco coir, peat moss, and composted bark, are often used in hydroponic systems to support plant roots, maintain moisture, and improve aeration. The use of organic substrates in hydroponics is particularly advantageous as they are renewable, biodegradable, and can enhance the sustainability of the system (Raviv et al., 2002).

Several studies have shown that hydroponic systems can significantly improve the growth and yield of leafy greens. For instance, Kyriacou et al. (2017) reported that hydroponically grown leafy vegetables exhibited superior growth performance and nutritional quality compared to their soil-grown counterparts. Additionally, research by Saha et al. (2016) indicated that the use of organic substrates in hydroponic systems could enhance the physical properties of the growing medium, thus supporting better root development and overall plant health.

The performance of hydroponically grown Purple Pechay using organic substrates is of particular interest due to the potential for maximizing yield and nutritional quality while

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maintaining sustainable practices. The integration of organic substrates into hydroponic systems can provide an effective medium for root growth, facilitating efficient nutrient uptake and water retention (Gruda, 2009). Moreover, the use of organic substrates aligns with the growing demand for environmentally friendly and sustainable agricultural practices (Raviv, 2010).

This study aims to evaluate the growth performance and yield of Purple Pechay cultivated in a hydroponic system utilizing different organic substrates. By examining these parameters, the research seeks to contribute to the understanding of sustainable hydroponic practices and their potential benefits for leafy vegetable production.

Objectives of the Study

The study was conducted to determine the performance of purple pechay under hydroponic culture using organic substrates. It specifically aimed to:

1. Determine the growth parameters of purple pechay as affected by the hydroponic substrates;
2. Assess the yield of purple pechay using hydroponic substrates;
3. Evaluate which among the organic substrates gave the best growth and yield to purple pechay.

(11) **Materials and Methods or Methodology** (This includes the materials, methods, duration & location of the study. Materials and Methods are headings used in Natural Science researches and methodology is Social Sciences. Presentation of materials may not necessarily be in separate heading but may be incorporated in the specific methods used. Justify paragraphs with main headings in caps.)

The study was conducted at the Organic Agriculture Project of CAPSU Burias, Mambusao, Capiz, Philippines. The experiment was laid out using the Completely Randomized Design, replicated three times. The treatments consisted of different substrates: A – SNAP, B – CAFSA with chicken dung and molasses, C – CAFSA with cow manure without molasses, D – CAFSA with chicken dung without molasses, and E – CAFSA with hog manure without molasses.

Growth and yield parameters such as plant height, stem girth, number of leaves, width of leaves, length of leaf, length of root, weight of root, biomass, weight of marketable part, and number of infected parts were gathered and analyzed using the Statistical Tool for Agricultural Research (STAR) software, except for the descriptive data. Differences among treatment means were compared using the Least Significant Difference test (LSD).

(12) **Results and Discussions** (This part contains presentation of results (data) with reference to the table and figure, to be followed by discussion, with support of reviewed literature. Sequence shall be based on the objectives of the study. Tables are formatted without the vertical lines. Horizontal lines are set at the top and bottom portions as well as below table headings and above major subdivisions. Justify paragraphs with main headings in caps.)

Plant Height (cm)

The data in Figure 1 indicate that plant height of purple pechay was significantly influenced by the hydroponic substrates used. SNAP produced the tallest plants, with an average height of 31.8 cm. Conversely, CAFSA with chicken dung without molasses resulted in the shortest plant height, averaging 23.40 cm. This was statistically comparable to CAFSA with hog manure without molasses (24.85 cm), CAFSA with chicken dung and molasses (25.55 cm), and CAFSA with cow manure without molasses (26.53 cm).

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Stem Girth (cm)

The Analysis of Variance (ANOVA) showed that stem girth of purple pechay was not significantly affected by the different hydroponic substrates. As presented in Figure 2, the mean stem girth ranged from 4.33 cm to 4.96 cm.

Number of Leaves

ANOVA revealed that the number of leaves of purple pechay was not significantly affected by the hydroponic substrates used. Figure 3 illustrates that the mean number of leaves ranged from 7.37 to 9.57.

Width of Leaf (cm)

Figure 4 shows that the width of leaf of purple pechay was significantly influenced by the hydroponic substrates. The widest leaf was recorded in SNAP with an average width of 11 cm. In contrast, CAFSA with chicken dung without molasses resulted in the narrowest leaves, averaging 8.24 cm. This was comparable to CAFSA with hog manure without molasses (8.39 cm), CAFSA with cow manure without molasses (8.46 cm), and CAFSA with chicken dung and molasses (8.97 cm).

Length of Leaf (cm)

Figure 5 reveals that the length of leaf of purple pechay was significantly affected by the hydroponic substrates. Notably, purple pechay applied with SNAP had the longest leaf, with an average length of 24.28 cm, while CAFSA with chicken dung without molasses resulted in the shortest leaf, averaging 17.53 cm. This result was comparable to CAFSA with hog manure without molasses (18.29 cm), CAFSA with chicken dung and molasses (20 cm), and CAFSA with cow manure without molasses (20.10 cm).

Length of Root (cm)

The length of root of purple pechay was significantly influenced by the hydroponic substrates. The data in Figure 6 present that SNAP recorded the longest roots, with an average length of 28.66 cm, followed by CAFSA with cow manure without molasses (19.81 cm), which was comparable to CAFSA with chicken dung and molasses (18.81 cm). Conversely, CAFSA with chicken dung without molasses produced the shortest roots, averaging 15.33 cm.

Weight of Root (g)

ANOVA revealed that weight of root of purple pechay was not significantly affected by the hydroponic substrates. Figure 7 shows that the mean root weight ranged from 1.90 g to 2.97 g.

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Biomass (g)

The biomass of purple pechay was significantly influenced by the hydroponic substrates. Figure 8 indicate that SNAP had the highest biomass, with an average of 50.77 g. In contrast, CAFSA with chicken dung without molasses resulted in the lowest biomass, averaging 25.13 g. This result was comparable to CAFSA with hog manure without molasses (25.47 g), CAFSA with chicken dung and molasses (27.77 g), and CAFSA with cow manure without molasses (29.20 g).

Weight of Marketable Part (g)

The data in Figure 9 indicate that the weight of the marketable part of purple pechay was significantly affected by the hydroponic substrates. SNAP produced the heaviest marketable parts, with an average weight of 48 g. Conversely, CAFSA with chicken dung without molasses resulted in the lowest weight, averaging 22.27 g. This was comparable to CAFSA with hog manure without molasses (23.17 g), CAFSA with chicken dung and molasses (25.87 g), and CAFSA with cow manure without molasses (26.23 g).

Number of Infected Parts

The data in Figure 10 demonstrate that ANOVA found no significant effect of hydroponic substrates on the number of infected parts of purple pechay. The mean number of infected parts ranged from 7 to 9.67.

(13) Conclusions and Recommendations (This part shall be written in narrative form and arranged based on the objectives and findings. Conclusions are general statements answering the objectives. Recommendations are based on the conclusions. They may be split or combined based on their length, which means, conclusions and recommendations may be split under separate headings if they are long. Justify paragraphs with main headings in caps.)

Conclusions

Based on the results of the study, the following conclusions were drawn:

1. The growth of purple pechay, as indicated by plant height, stem girth, number of leaves, width and length of leaves, length of roots, and root weight, was influenced variably by the hydroponic substrates.
2. The yield of purple pechay, as indicated by biomass and weight of marketable parts, was significantly influenced by the hydroponic substrates.
3. Among the organic substrates, CAFSA with cow manure without molasses performed better with higher plant height wider leaves, longer leaves, and longer roots, heavier biomass and weight of marketable part.

Recommendations

Based on the results of the study focusing on organic substrates, the following recommendations are forwarded:

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1. Given its superior performance in plant height, leaf width, leaf length, root length, biomass, and marketable weight, it is recommended to use CAFSA with cow manure without molasses for cultivating purple pechay.
2. Explore and test various combinations and formulations of CAFSA with different types of organic manures, such as chicken dung, hog manure, and cow manure, to determine their impact on the growth and yield of purple pechay. Adjusting the proportions and additives may yield better results.
3. Investigate the effects of incorporating molasses into CAFSA with different types of organic manures. Although CAFSA with chicken dung and molasses showed some benefits, further research is needed to fully understand its impact on plant growth and yield.
4. Assess the nutrient profiles of the various organic substrates used in the study to ensure they meet the nutritional requirements of purple pechay. This will help in identifying any deficiencies or imbalances that may affect plant growth and yield.
5. Encourage farmers to use locally available organic materials, which can reduce costs and environmental impact while improving soil health and crop productivity.

(14) Output (Indicate the specific products, processes or services which the project has produced; quantify when possible.)

(15) Technology Transfer Activity Conducted (Indicate the specific products, activity design, processes or services transferred as extension activities)

(16) Economic Analysis (This should present the direct cost of project implementation, expenses in the production of technology, revenues generated from technology produced, return on investment (ROI) and payback period.

(17) Acknowledgments (Include only important persons and institutions with involvement in funding and conduct of the research. Justify paragraphs with main headings in caps.)

(18) References (It should include only works mentioned in the text and should be arranged alphabetically. Citations shall not be less than 13 entries. This must follow the APA latest edition format, in hanging indentation with 12-point space in between.)

- Gruda, N. (2009). "Do soilless culture systems have an influence on product quality of vegetables?" *Journal of Applied Botany and Food Quality*, 82(2), 141-147.
- Jones, J. B. (2005). *Hydroponics: A practical guide for the soilless grower*. CRC Press.
- Kyriacou, M. C., Roupheal, Y., Colla, G., Zrenner, R., & Schwarz, D. (2017). "Vegetable grafting: A toolbox for securing yield stability under multiple stress conditions." *Frontiers in Plant Science*, 8, 225.
- Raviv, M., Wallach, R., Silber, A., & Bar-Tal, A. (2002). "Substrates and their analysis." In: *Hydroponic production of vegetables and ornamentals*. Embryo Publications, Athens, Greece, pp. 25-101.
- Raviv, M. (2010). "Sustainability of organic soilless growing media for greenhouse crops." *Acta Horticulturae*, 883, 341-347.
- Resh, H. M. (2012). *Hydroponic food production: A definitive guidebook for the advanced home gardener and the commercial hydroponic grower*. CRC Press.

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Saha, S., Monroe, A., & Day, M. R. (2016). "Growth, yield, plant quality and nutrition of basil (*Ocimum basilicum* L.) under soilless agricultural systems." *Annals of Agricultural Sciences*, 61(2), 181-186.

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APPENDICES

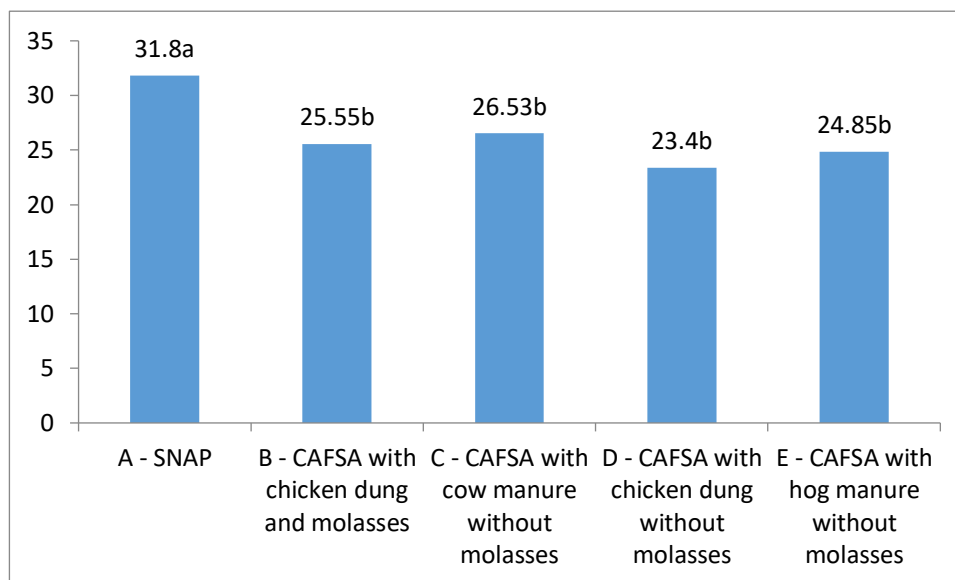


Figure 1. Plant height (cm) of purple pechay under hydroponic culture as affected by the different organic substrates.

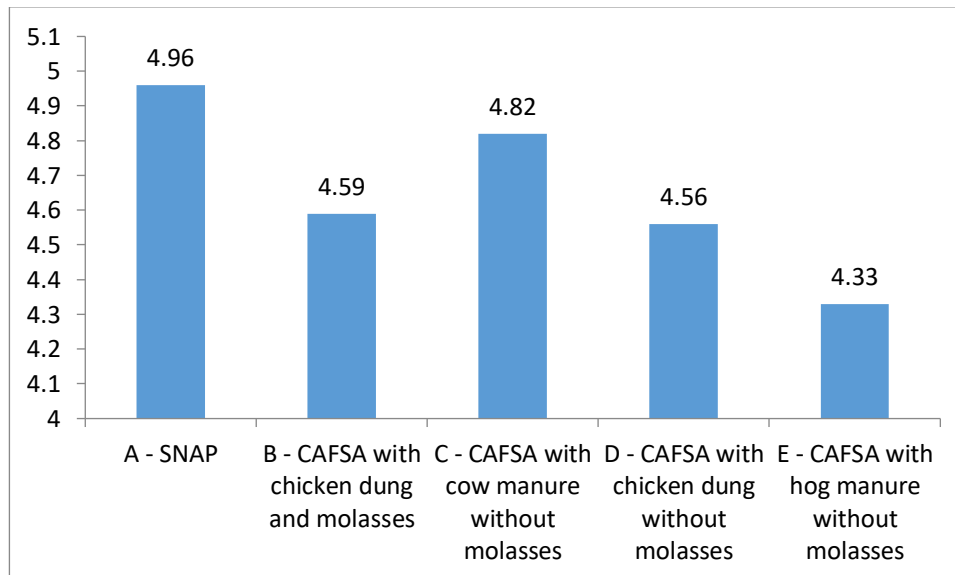


Figure 2. Stem girth (cm) of purple pechay under hydroponic culture using organic substrate.

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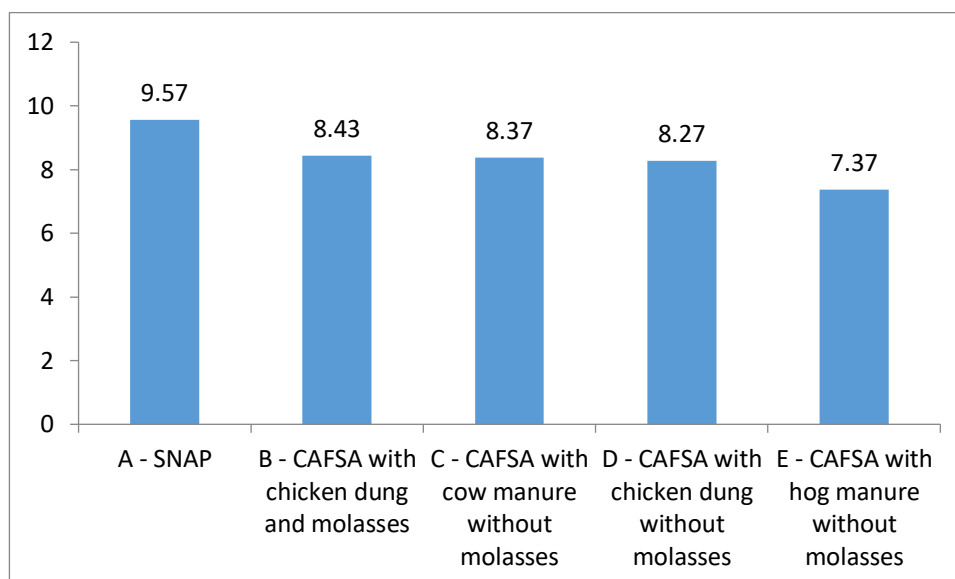


Figure 3. Number of leaves of purple pechay under hydroponic culture using organic substrate.

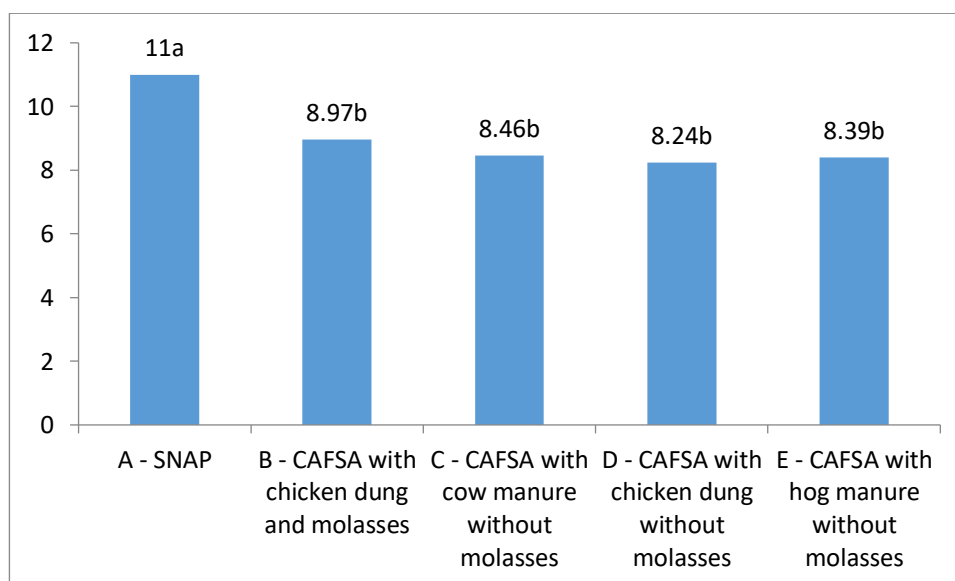


Figure 4. Width of leaf (cm) of purple pechay under hydroponic culture as affected by the different organic substrates.

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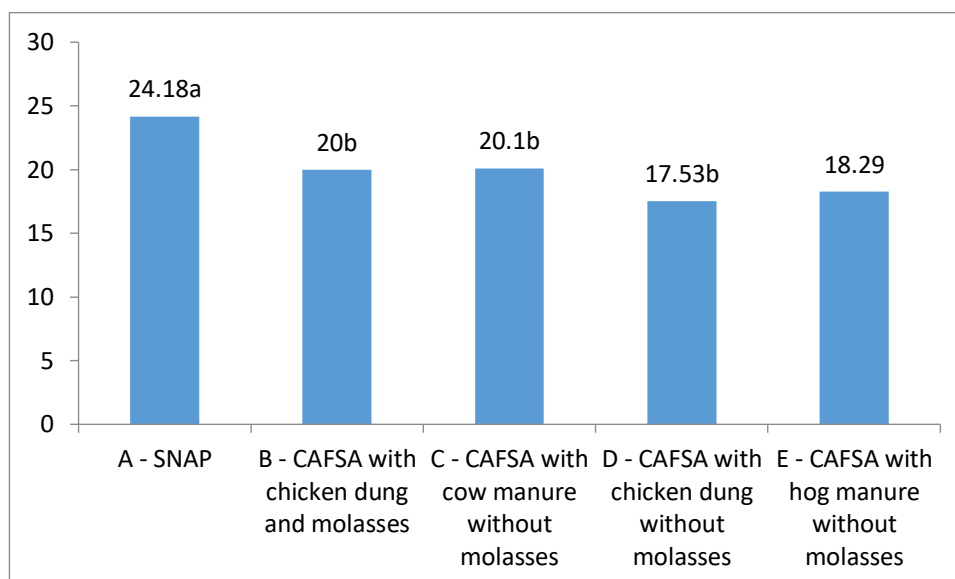


Figure 5. Length of leaf (cm) of purple pechay under hydroponic culture as affected by the different organic substrates.

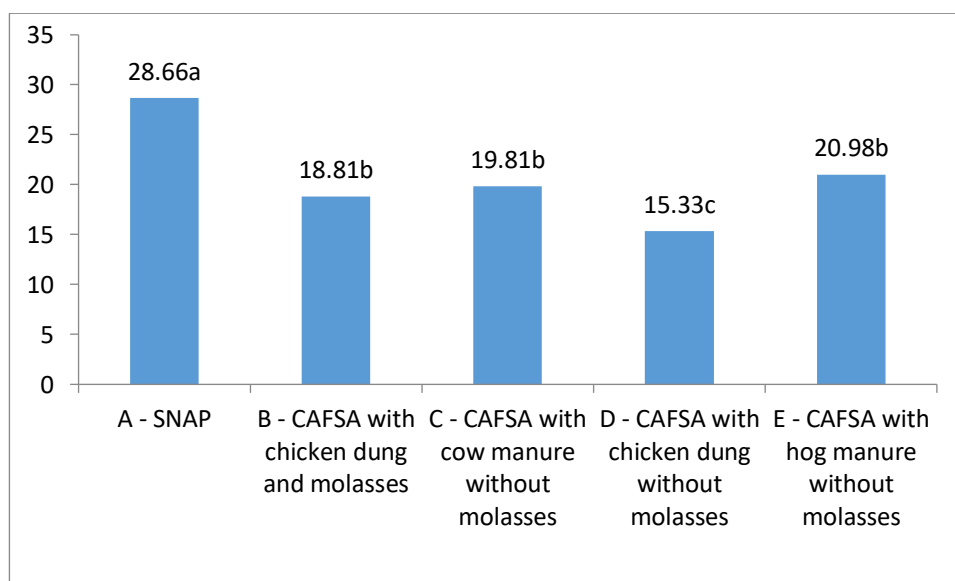


Figure 6. Length of root (cm) of purple pechay under hydroponic culture as affected by the different organic substrates.

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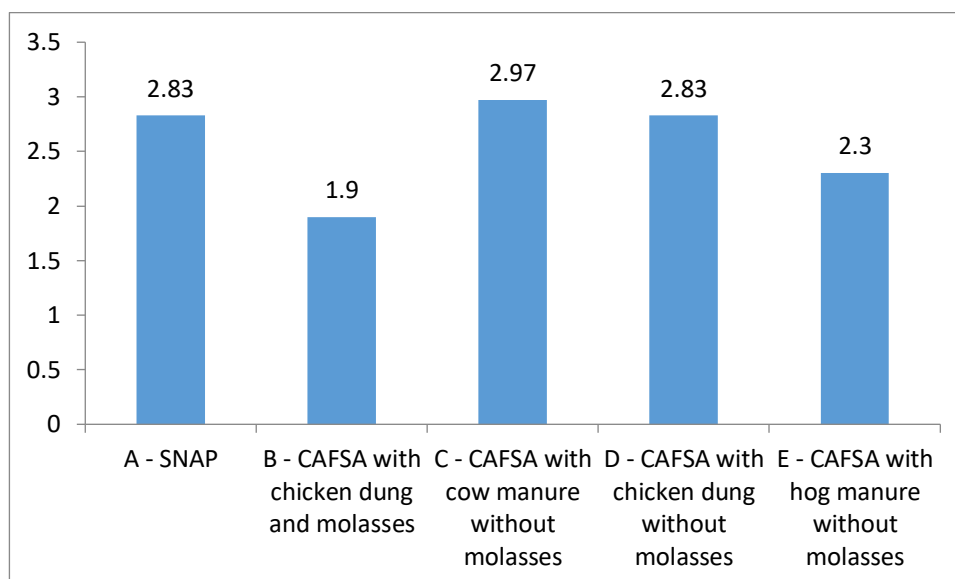


Figure 7. Weight of roots (g) of purple pechay under hydroponic culture using organic substrate.

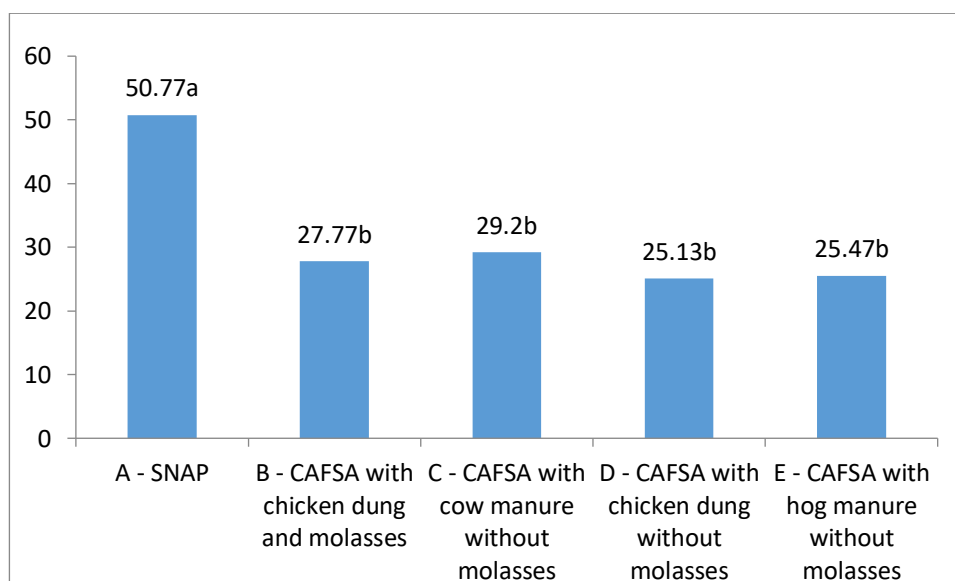


Figure 8. Biomass (g) of purple pechay under hydroponic culture as affected by the different organic substrates.

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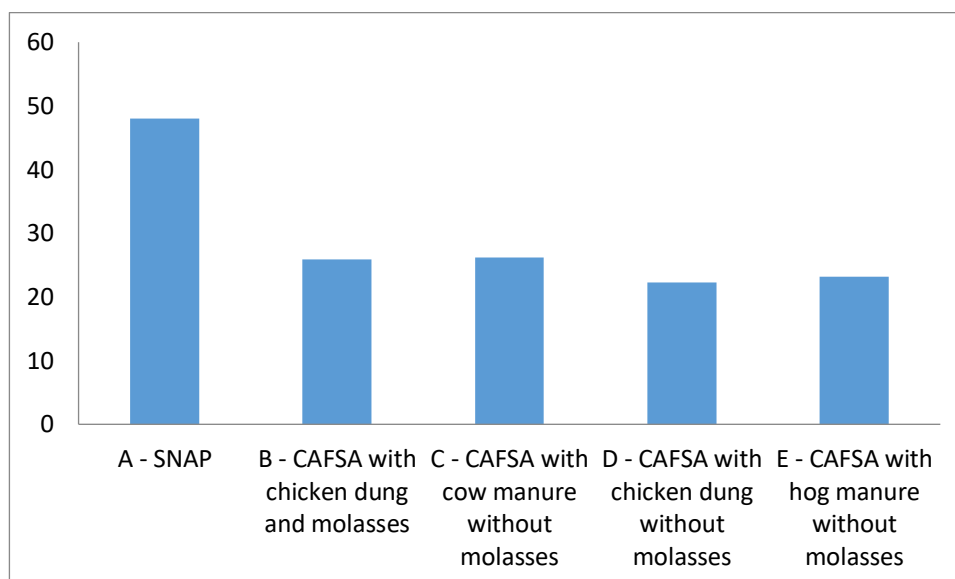


Figure 9. Weight of marketable part (g) of purple pechay under hydroponic culture as affected by the different organic substrates.

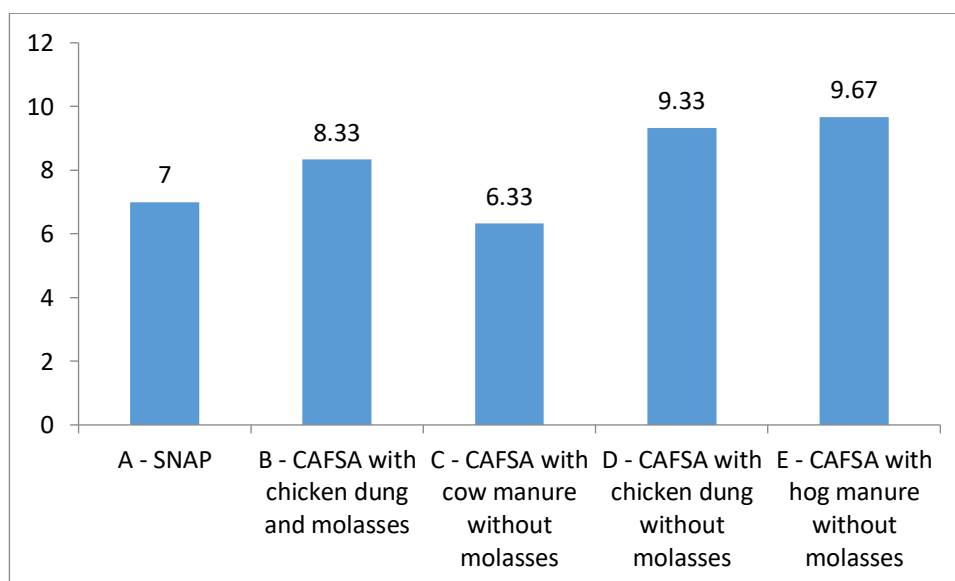


Figure 10. Number of infected part of purple pechay under hydroponic culture using organic substrate.

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Table 1. Plant height (cm) of purple pechay under hydroponic culture using organic substrate.

Plant height**					
TREATMENT	REPLICATION			TOTAL	MEAN
	I	II	III		
A	31.8	30.1	33.5	95.4	31.8a
B	23.05	25.59	28	76.64	25.55b
C	26.95	23.35	29.3	79.6	26.53 b
D	23.19	24.1	22.9	70.19	23.40 b
E	23.5	24.24	26.82	74.56	24.85 b
TOTAL	128.49	127.38	140.52		
MEAN	25.698	25.476	28.104		

Table 2. Stem girth (cm) of purple pechay under hydroponic culture using organic substrates.

Stem girth^{ns}					
TREATMENT	REPLICATION			TOTAL	MEAN
	I	II	III		
A	4.57	4.72	5.6	14.89	4.96
B	4.62	4.39	4.75	13.76	4.59
C	4.59	4.62	5.24	14.45	4.82
D	4.81	4.76	4.1	13.67	4.56
E	4.14	4.41	4.45	13	4.33
TOTAL	22.73	22.9	24.14		
MEAN	4.55	4.58	4.83		

Table 3. Number of leaves of purple pechay under hydroponic culture using organic substrate.

Number of Leaves^{ns}					
TREATMENT	REPLICATION			TOTAL	MEAN
	I	II	III		
A	8.6	8.6	11.5	28.7	9.57
B	9	8.3	8	25.3	8.43
C	8.1	8.2	8.8	25.1	8.37
D	7.6	8.9	8.3	24.8	8.27
E	7.2	7.3	7.6	22.1	7.37
TOTAL	40.5	41.3	44.2		
MEAN	8.1	8.26	8.84		

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Table 4. Width of leaf (cm) of purple pechay under hydroponic culture using organic substrate.

Width of Leaf*					
TREATMENT	REPLICATION			TOTAL	MEAN
	I	II	III		
A	10.9	10.44	11.65	32.99	11.00a
B	8.43	8.4	10.09	26.92	8.97b
C	8.57	7.31	9.5	25.38	8.46b
D	8.58	8.65	7.5	24.73	8.24b
E	7.85	8.13	9.19	25.17	8.39b
TOTAL	44.33	42.93	47.93		
MEAN	8.866	8.586	9.586		

Table 5. Length of leaf (cm) of purple pechay under hydroponic culture using organic substrate.

Length of Leaf*					
TREATMENT	REPLICATION			TOTAL	MEAN
	I	II	III		
A	23.3	24.45	24.8	72.55	24.18a
B	17.9	18.99	23.1	59.99	20.00b
C	21.2	17.5	21.6	60.3	20.10b
D	18.3	17.8	16.5	52.6	17.53b
E	16.4	17.69	20.77	54.86	18.29
TOTAL	97.1	96.43	106.77		
MEAN	19.42	19.286	21.354		

Table 6. Length of roots (cm) of purple pechay under hydroponic culture using organic substrate.

Length of Roots**					
TREATMENT	REPLICATION			TOTAL	MEAN
	I	II	III		
A	29.05	28.3	28.63	85.98	28.66a
B	18.3	18	20.12	56.42	18.81b
C	19.11	20.72	19.6	59.43	19.81b
D	16.68	15	14.3	45.98	15.33c
E	20.05	19.3	23.6	62.95	20.98b
TOTAL	103.19	101.32	106.25		
MEAN	20.638	20.264	21.25		

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Table 7. Weight of roots (cm) of purple pechay under hydroponic culture using organic substrate.

Weight of roots^{ns}					
TREATMENT	REPLICATION			TOTAL	MEAN
	I	II	III		
A	2.3	2.6	3.6	8.5	2.83
B	1.3	2.4	2	5.7	1.90
C	1.6	4.3	3	8.9	2.97
D	4	1.3	3.2	8.5	2.83
E	2.3	2.3	2.3	6.9	2.30
TOTAL	11.5	12.9	14.1		
MEAN	2.3	2.58	2.82		

Table 8. Biomass (g) of purple pechay under hydroponic culture using organic substrate.

Biomass*					
TREATMENT	REPLICATION			TOTAL	MEAN
	I	II	III		
A	38.3	39.7	74.3	152.3	50.77a
B	22.1	27.1	34.1	83.3	27.77b
C	25.3	26.1	36.2	87.6	29.20b
D	24.8	24.3	26.3	75.4	25.13b
E	22.6	25.1	28.7	76.4	25.47b
TOTAL	133.1	142.3	199.6		
MEAN	26.62	28.46	39.92		

Table 9. Weight of marketable part (g) of purple pechay under hydroponic culture using organic substrate.

Weight of Marketable Part*					
TREATMENT	REPLICATION			TOTAL	MEAN
	I	II	III		
A	36	37.3	70.7	144	48.00a
B	20.8	24.7	32.1	77.6	25.87b
C	23.7	21.8	33.2	78.7	26.23b
D	20.8	23	23	66.8	22.27b
E	20.3	22.8	26.4	69.5	23.17b
TOTAL	121.6	129.6	185.4		
MEAN	24.32	25.92	37.08		

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Table 10. Number of infected part Of n purple pechay under hydroponic culture using organic substrate.

Number of Infected Part^{ns}					
TREATMENT	REPLICATION			TOTAL	MEAN
	I	II	III		
A	6	6	9	21	7.00
B	10	6	9	25	8.33
C	8	5	6	19	6.33
D	8	10	10	28	9.33
E	13	8	8	29	9.67
TOTAL	45	35	42		
MEAN	9	7	8.4		